

Seminar 2 - DSA

18.03.2026

1. a) $m^2 \in O(n^3)$ T.
 b) $m^3 \in O(n^2)$ F.
 c) $2^{n+1} \in \Theta(2^n)$ T $2^{n+1} = 2 \cdot 2^n$
 d) $2^{2n} \in \Theta(2^n)$ F $2^{2n} = 2^n \cdot 2^n$
 e) $m^2 \in \Theta(n^3)$ F
 f) $2^n \in O(n!)$ T.
 g) $\log_{10} n \in \Theta(\log_2 n)$ T
 $\hookrightarrow \frac{\log_2 n}{\log_{210}} \leftarrow \text{ct.}$

h) $O(n) + \underline{\underline{\Theta(n^2)}} = \underline{\underline{\Theta(n^2)}}$ T.

i) $\Theta(n) + \underline{\underline{O(n^2)}} = \underline{\underline{O(n^2)}}$ T.

two $\neq$ pieces of code.

$\Theta(n) + \underline{\underline{O(n^2)}} = \underline{\underline{\Omega(n)}}$

$\Theta(1)$
BC

$\Theta(n^2)$
WC

lower bound.

- j) $O(n) + \Theta(n) = O(n)$ T by definition,
 but we will use $\Theta(n)$.

$T(n) = 3n^2 + 5n \cdot \log n + 1000 \rightarrow$ "abstract view" expression.

leading term

$\downarrow$
 Θ

(a) $O(n^5)$

(b) $\Omega(n^3)$

(c) $\Theta(n^3)$

(d) $O(n^2)$

(e) $\Theta(n^2)$

$\cancel{f) \Theta(n \cdot \log n)}$

k) $O(n) + O(n^2) = O(n^2)$ T.

l) $(m+n)^2 \in O(n^2 + m^2)$ T.

$\hookrightarrow m^2 + 2nm + n^2$. Assume $m > n \Rightarrow (m+n)^2 < 4m^2$
 $m > n \Rightarrow (m+n)^2 < 4m^2$

$(m+n)^2$ upper.

$O(m+n^2)$ not because m^2 is bigger $\Rightarrow O(m^2)$??

u^4 in complexities: max. out of the two values.
Graph algorithms $\rightarrow$ complexities with u^4 because idk how many nodes/edges!!

m) $3^m \in O(2^m)$ F.

n) $\log_2 3^m \in O(\log_2 2^m)$ T

$(\underbrace{m \cdot \log_2 3}_{ct} \in \underbrace{m \cdot \log_2 2}_{ct})$

②. Complexity of each search/sort algorithm.

! Selection sort does not have BC, WC, AC...

! Extra space complexity = how much space needed besides that array mainly used by alg.

• $BC = WC = AC \Rightarrow TC = \Theta$
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• $BC \neq WC \neq AC \Rightarrow TC = WC \Rightarrow O(sth)$

③a, ③b, ...

3a
Computational ways of computing complexities:

(K1) for loop $\rightarrow$ sum! $\sum_{i=1}^m \log m = \log m + \log m + \dots + \log m = \Theta(m \cdot \log m)$
how many instructions

(K2) write the iterations separately:

$i=1$ $\log m$ instr.

$i=2$ $\log m$ instr

$\vdots$

$i=m$ $\log m$

$\Rightarrow \Theta(m \cdot \log m)$

numbers from $[0, N]$. Sort S based on the keys

$$\sum_{i=1}^m \log i = \log 1 + \log 2 + \dots + \log m = \log(1 \cdot 2 \cdot \dots \cdot m) = \log(m!)$$

$\Rightarrow$ Stirling's approximation $\Rightarrow \Theta(m \log m)$

4a) $\Theta(m) \rightarrow$ "found" flag not used anywhere.

4b) BC: $\Theta(1)$ MIN. nr. of steps
 WC: $\Theta(m)$ MAX. nr. of steps

AC: $\sum_{I \in D} P(I) \cdot E(I) = \sum_{i=1}^m \frac{1}{n+1} \cdot i + \frac{m}{n+1}$

$\swarrow$ $n+1$ cases

$\nearrow$ in while $\Rightarrow$ st. nr. of instructions (once)
 $\nearrow$ repeat while loop n times.

$$= \frac{(1 + \dots + m) + m}{n+1} = \frac{\frac{n(n+1)}{2} + \frac{2}{n}}{n+1} = \frac{n(n+1) + 2m}{2n(n+1)} = \frac{(n+1)(n+2) - 1}{n+1} \in \Theta(n) \Rightarrow TC: O(n)$$

$\Rightarrow TC: O(n)$

5) $\sum_{i=1}^m \sum_{j=1}^{x_i} 1 = \sum_{i=1}^m x_i = S \Rightarrow \Theta(S)$

$\downarrow$ first for loop $\downarrow$ second for loop $\uparrow$ sum of elements in the array

can be $\boxed{x_i = 0}$

$x_i = \begin{cases} 1 & \text{if } i \text{ is a pf. square} \\ 0 & \text{otherwise} \end{cases}$

(particular array ex.)

$\Rightarrow S = \sqrt{m} \Rightarrow$ complexity is not $\Theta(\sqrt{m})$ because of the **for loop** with m elements.

$\Rightarrow \Theta(s+m)$ $T(x, m) \in \Theta(\max\{m, s\}) = \Theta(m+s)$

MEANS $\swarrow \searrow$
 $\swarrow$ "++" because idk which one is max.

$\begin{cases} \Theta(s) \rightarrow \text{for some} \\ \Theta(m) \rightarrow \text{for others} \end{cases}$, depending on which is max.

④ While with flags. $\Rightarrow \exists BC, AC, WC$

BC: $\Theta(m)$

WC: while executed many times, flags are $\begin{cases} \text{true} \\ \text{false} \end{cases}$?

$$T(n) = n + \sum_{i=1}^{n-1} \sum_{j=i}^n 1 = n + \sum_{i=1}^{n-1} (n-i+1) = n + \sum_{i=1}^{n-1} n - \sum_{i=1}^{n-1} i + \sum_{i=1}^{n-1} 1$$

$\downarrow$ 1st while loop $\downarrow$ 2nd while loop

$$= n + \sum_{i=1}^{n-1} n - \frac{n(n-1)}{2} + \frac{n-1}{2}$$

$$= \frac{2n + 2n^2 - 2n - n^2 + n + 2n - 2}{2} = \frac{n^2 + n - 2}{2} \in \Theta(n^2)$$

AC: For the inner while, for a fixed value of i .

$$\sum_{I \in \Omega} P(I) \cdot E(I)$$

probability

complexity

probability depending on nr. of executions:

$$P(I) = \frac{1}{n-i+1}$$

$$T(m, i) = \frac{1}{n-i+1} \cdot 1 + \frac{1}{n-i+1} \cdot 2 + \frac{1}{n-i+1} \cdot 3 + \dots + \frac{1}{n-i+1} (m-i+1)$$

$\downarrow$ nr. of instructions $\downarrow$ max. nr. of repetitions $\downarrow$ while once $\downarrow$ while twice

$$= \frac{1}{n-i+1} (1 + 2 + 3 + \dots + m-i+1)$$

$$= \frac{1}{n-i+1} \cdot \frac{(m-i+1)(m-i+2)}{2}$$

$$= \frac{n-i+2}{2}$$

average nr. of instructions for fixed m, i .

AC: For the outer while:

$$P(i) = \frac{1}{m-1}$$

Nr. of executions:

nr. of executions:

$$\begin{aligned}
 1 &\Rightarrow \frac{1}{n-1} (1 + T(n,1)) \\
 2 &\Rightarrow \frac{1}{n-1} (2 + T(n,1) + T(n,2)) \\
 3 &\Rightarrow \frac{1}{n-1} (3 + T(n,1) + T(n,2) + T(n,3)) \\
 &\vdots \\
 n-1 &\Rightarrow \frac{1}{n-1} ((n-1) + T(n,1) + T(n,2) + \dots + T(n, n-1))
 \end{aligned}$$

$i=1$

(*)

how many times
the while
is executed.

probability for
every case for
executing the
while loop once/twice/...

↓

$$\begin{aligned}
 &\frac{1}{n-1} \left(\frac{(n-1)n}{2} + (n-1) \cdot T(n,1) + (n-2) \cdot T(n,2) + \dots + T(n,3) \cdot (n-3) + \dots \right. \\
 &\quad \left. + \dots + 1 \cdot T(n, n-1) \right) = \\
 &= \frac{1}{n-1} \cdot \left(\frac{n^2 - n}{2} + \sum_{k=1}^{n-1} (n-k) \cdot T(n, k) \right) \\
 &= \frac{1}{n-1} \cdot \left(\frac{n^2 - n}{2} + \sum_{k=1}^{n-1} (n-k) \cdot \frac{n-k+2}{2} \right) \in \Theta(n^2).
 \end{aligned}$$

BC: $\Theta(n)$

WC: $\Theta(n^2)$

AC: $\Theta(n^2)$

$\Rightarrow TC: O(n^2)$.

how much time
space needed
besides that array
↑ used by
alg.

E.S.

Complexity:	TIME				E.S.
	BC	WC	AC	Total	
Linear Search	$\Theta(1)$	$\Theta(n)$	$\Theta(n)$	$O(n)$	$\Theta(1)$
Binary Search	$\Theta(1)$	$\Theta(\log_2 n)$	$\Theta(\log_2 n)$	$O(\log_2 n)$	$\Theta(1)$
Selection Sort	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(1)$ - in place.
Insertion Sort	$\Theta(n)$	$\Theta(n^2)$	$\Theta(n^2)$	$O(n^2)$	$\Theta(n)$ - in place.
Bubble Sort	$\Theta(n)$	$\Theta(n^2)$	$\Theta(n^2)$	$O(n^2)$	$\Theta(1)$ - in place.
Quick Sort	$\Theta(n \log_2 n)$	$\Theta(n^2)$	$\Theta(n \log_2 n)$	$O(n^2)$	$\Theta(1)$ - in place.
Merge Sort	$\Theta(n \log_2 n)$	$\Theta(n \log_2 n)$	$\Theta(n \log_2 n)$	$\Theta(n \log_2 n)$	$\Theta(n)$ - out of place.